

ForecastingCore — Technical Specification

Pipeline Overview

ForecastingCore is a Python ML library that orchestrates multi-model time-series forecasting. The pipeline consists of: data loading, feature engineering, model training (per SKU), evaluation, and forecast generation. All models are sklearn-compatible and support walk-forward cross-validation.

Feature Engineering

Calendar features include day-of-week, month, week-of-year, and Colombian holiday distances (Easter, Christmas, Semana Santa). Lag features cover configurable look-back windows. Rolling statistics include mean, std, min, max and trend via polynomial fit.

ML Models (LightGBM / XGBoost)

Gradient boosting models trained per SKU. Hyperparameter tuning via Optuna (optional). Walk-forward validation with expanding window. Residuals stored for prediction interval construction at inference time.

Statistical Models (ARIMA, Prophet, ETS, Croston)

Statistical models bypass feature engineering and operate on raw time series. ARIMA supports seasonal differencing. Prophet handles multiple seasonalities and holiday effects. ETS covers trend/seasonal components. Croston targets intermittent demand patterns.

Evaluation Metrics

MAE: mean absolute error. RMSE: root mean squared error. WAPE: weighted absolute percentage error (robust to zeroes). Bias: directional error indicating systematic over/under-forecasting. All metrics computed out-of-sample on the held-out test fold.

Inventory Advisor

Safety stock = $Z * \sigma * \sqrt{\text{lead_time}}$. Reorder point = $\mu * \text{lead_time} + \text{safety_stock}$. Stockout risk estimated from forecast error distribution. Holding cost proportional to average inventory level.